How-To Library

Informatica

Disaster Recovery for Data Engineering Integration 10.4 on Microsoft Azure

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Abstract

You can configure disaster recovery to minimize business disruptions. This article describes how to implement disaster recovery and high availability for an Informatica® Data Engineering Integration implementation on Microsoft Azure.

Supported Versions

- Informatica Data Engineering Integration 10.4.0

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Overview

Implement a disaster recovery plan to minimize data loss and business disruptions, and to restore the system to optimal performance.

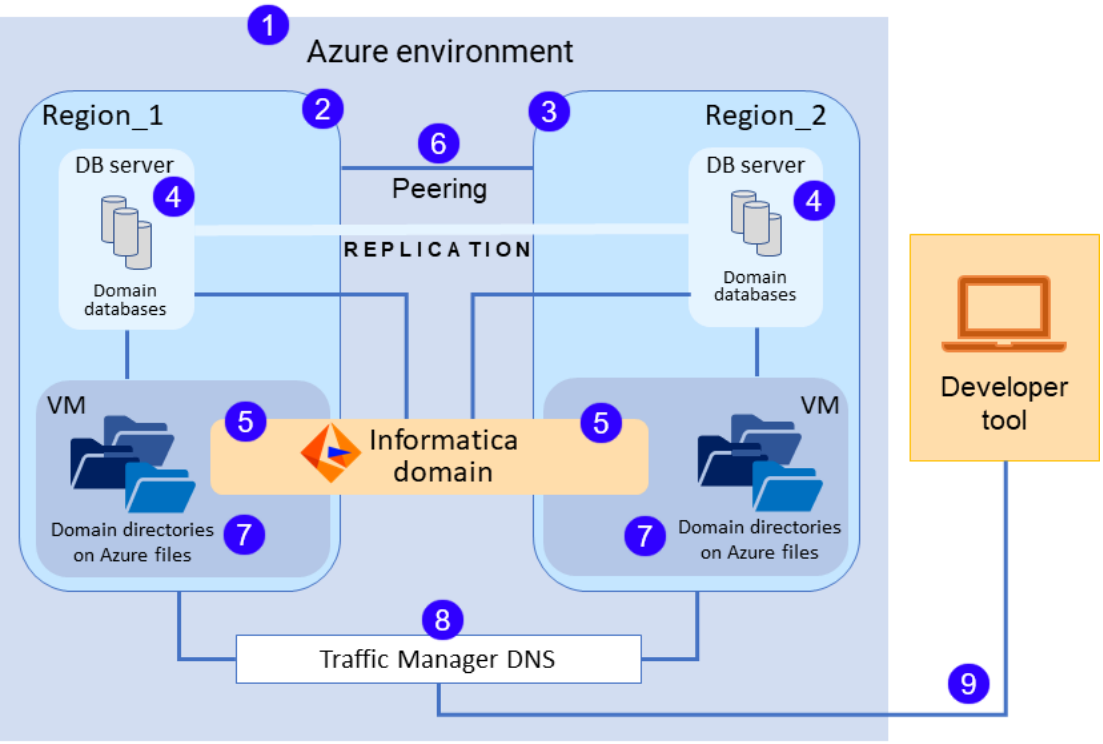
When you set up disaster recovery, you create a primary node and a secondary node on virtual machines (VMs) in virtual networks (vnets) in separate Azure geographic regions.

Informatica services stores metadata in Azure SQL server databases. Set up replication on database servers in different regions, so that when the database in one region goes down, the database in the other region comes up. This failover capability makes the domain highly available.

You also enable the Developer tool to maintain a domain connection through Azure Traffic Manager.

Architecture

The following image shows the architecture of a highly available implementation of Data Engineering Integration in an Azure environment:



The numbered elements in the diagram correspond to the following descriptions:

1. Azure account.
2. Primary region.
3. Secondary region.
4. Azure SQL databases for the Informatica domain, replicated across regions.
5. Informatica domain nodes in each region. The Informatica domain spans the VMs in each region.
6. Peering between regions.
7. Directories in Azure files for domain sources, targets, logs, and more.
8. Azure Traffic Manager DNS.
9. Developer tool uses the Traffic Manager DNS name as the connection node.

Note: The diagram and the following procedure use the terms Region_1 and Region_2 to refer to the primary and secondary Azure geographical regions where you implement the solution. In addition, the procedure uses placeholder names to refer to system elements. For example, Step 1 says to create a SQL database server called `infasrvrprimaryrg`. You can choose your own names for system elements.

Prerequisites

Before you begin to implement disaster recovery, verify the following Azure prerequisites:

- You have a Microsoft Azure subscription.

- You have access and permissions to create the following resources on the Azure platform:
  - Virtual networks (vnets) in separate geographic regions. The implementation of a disaster recovery strategy requires separate Azure regions to host the primary and secondary domain nodes.
    For help identifying the regions for the primary and secondary nodes. see the Azure documentation.
  - Network security group
  - Virtual machines (VMs)
  - SQL databases
  - A sufficient number of CPU cores based on the instance types in both of the regions where you plan to deploy Data Engineering Integration.

Step 1. Set Up SQL Databases in the Primary Region

Set up database users and schemas for an Azure SQL database to host the Model repository.

Create and configure the SQL database in an Azure vnet in the primary region, Region_1.

Log in to the Azure portal to perform these steps to create the primary database and its users:

1. Create an Azure SQL database server `infasrvrprimaryrg` in Region_1.
2. In `infasrvrprimaryrg`, create an SQL database `infadbprimaryRG`.
3. Follow Azure documentation to perform the following tasks in the master database:
   - Create a database user account.
   - Create users for the Informatica domain and the Model repository.

Step 2. Create a VM and ADLS Gen2 Storage

Create and mount a storage account on the primary VM. Then create directories for the sources, logs, and other files that the Informatica domain uses.

1. In Region_1, create a Linux VM to use as the Informatica domain host machine.
2. In the Azure portal, create an ADLS Gen2 storage account.
3. Mount the storage account file system on each VM in Region_1.
   Follow the instructions in the Azure documentation.
4. In the file system that you created, create the directories for sources, logs, and other files that Informatica reads from or writes to. The files include, but are not limited to, the following:
   - Sources and targets
   - Domain logs
   - Domain backups
   - Data Integration Service process logs

Step 3. Deploy Informatica Data Engineering Integration on the Primary VM

Deploy the Informatica domain on the Region_1 primary VM. Use the database logins you created in "Step 1. Set Up SQL Databases and the Informatica Domain in the Primary Region."

Step 4. Create a Secondary Node in Region_2

Establish a secondary node in a separate geographical region, to use as the failover domain.

Create the secondary database and its users for the Informatica domain.

1. In Region_2, create a VM.
2. On the Region_2 VM, mount the storage that you created in "Step 2. Create a VM and ADLS Gen2 Storage."
3. Create an Azure SQL database server `infasrvrsecondaryrg` in Region_2.
4. In `infasrvrsecondaryrg`, create an SQL database `infadbsecondaryRG`.

Step 5. Configure Peering

Configure peering between the two regions that you created. Peering enables VMs in two regions to behave as if they are in the same region.

1. Log into the primary VM in Region_1 and select **Peerings** from the options on the left.
2. Select **Add Peering**.
   The **Add Peering** configuration panel opens.
3. Configure the following settings:

Property	Description
Name of the peering from <Region_1> to remote virtual network	Type a name for the peering configuration that Region_1 uses to establish peering with Region_2. For example, `<Region_1>to<Region2>`
Virtual network deployment model	Use the default setting, **Resource manager**.
Subscription	Select the subscription to use for the failover system.
Name of the peering from remote virtual network to <Region_1>	Type a name for the peering configuration that Region_2 uses to establish peering with Region_1. For example, `<Region_2>to<Region_1>`

4. Click **OK**.

Step 6. Create a Failover Group

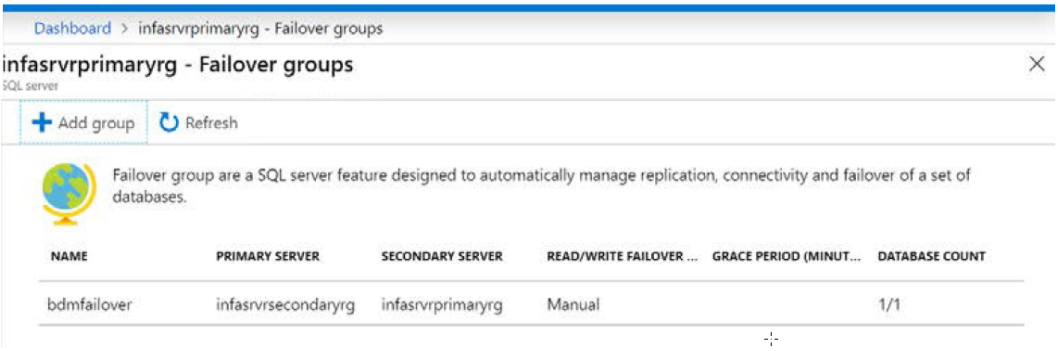
Create a failover group to act as an endpoint of domain requests to the database.

1. In the primary vnet, open the database server dashboard.
2. Create a failover group.

The following table contains the properties to configure:

Property	Description	Value
Failover group name	Name of the group. **Note:** Azure appends the domain string .database.windows.net to the name. For example, `bdmfailover.database.windows.net`	(Required input)
Secondary server	Click the arrow symbol **>** to display the database servers in other regions in your account.	(Select the Region_2 database server)
Read/Write failover policy	Policy to apply to the group	Manual

3. Click **Create**.
The image below shows the failover group listed on the dashboard:



Step 7. Create Logins and Map Database Users

In Region_2, create logins for, and map Region_1 users to, the Region_2 database.

1. Create database logins in Region_2. Create the login by mapping the Region_1 database user SID to the Region_2 database server:
   a. In the Region_2 database server, run the following query to fetch the SID:
      ```
      SELECT sid FROM sys.syslogins WHERE name = 'sqlLoginName'
      ```
      Use the SID in the next step.
   b. In the Region_2 master database, create a database login and a user account for that login:
      ```
      CREATE LOGIN <SQL login name> WITH PASSWORD = 'Testersql@123', SID = <SID>;
      ```
   Repeat this step for each of the database logins that you created in the Region_1 database.
   Note: because the database in region one is replicated to Region_2, it is not necessary to create database users.
2. Map the users in the Region_1 SQL database with the users replicated in the Region_2 SQL database.

Step 8. Configure High Availability

Configure high availability between the gateway nodes.

When the domain has multiple gateway nodes, the master gateway can fail over to the secondary node. Configure multiple gateway nodes to prevent domain shutdown when the master gateway node is unavailable.

1. In Region 2, run the Informatica installer and join the domain that you created in Region 1.

   Note: Be sure that you are creating a node as a member of the domain you installed in Region_1. For more information about joining an existing domain, see the *Informatica Installation Guide*.

2. Designate a node on the Region_2 Informatica instance as a gateway.

   Note: By default, the node on the Region_1 VM serves as the master gateway.

3. Configure the Data Integration Service and Model Repository Service on node 1 as the primary node, and designate node 2 as the backup node.

4. Use infasetup to change the JDBC connection endpoint to the DNS name of the failover group.
   For example, the following command contains the string `jdbc:informatica:sqlserver://<failover database name>.windows.net:1433` as the failover group DNS name:

   ```
   ./infasetup.sh updategatewaynode -cs "jdbc:informatica:sqlserver://<failover database name>.windows.net:1433;database=<primary database name>;encrypt=true;trustServerCertificate=false;hostNameInCertificate=*.database.windows.net;loginTimeout=30;EncryptionMethod=SSL;ValidateServerCertificate=false" -du <database user name> -dp <database user password> -dt MSSQLServer -dn <domain name>
   ```

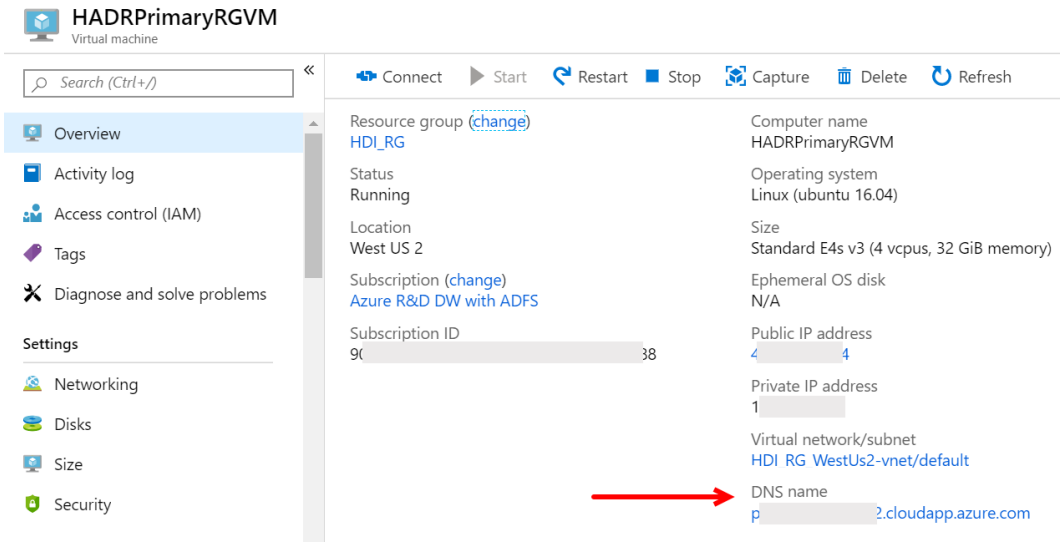
For more information about setting up high availability, see the *Informatica Administrator Guide*.

Step 9. Configure Azure Traffic Manager

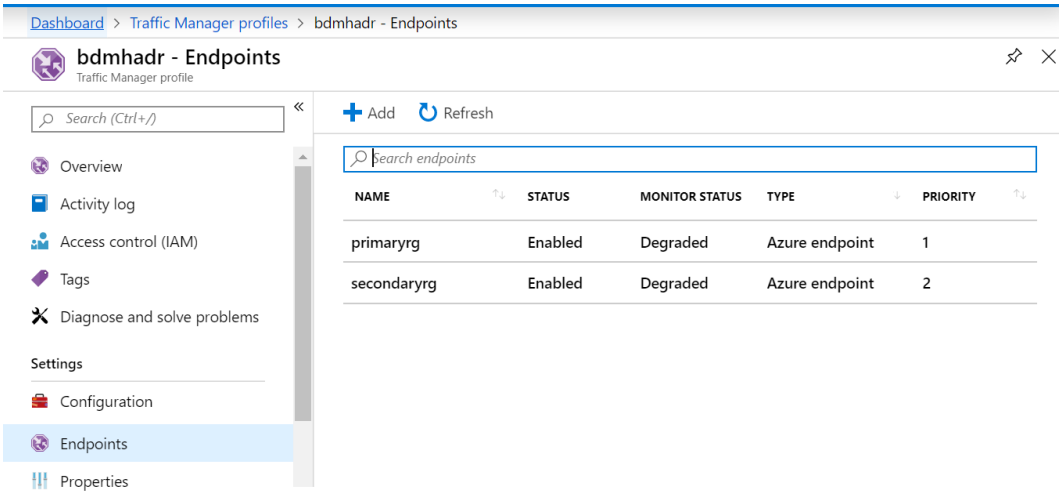
Configure Azure Traffic Manager to provide a dynamic endpoint for the Developer tool connection.

The Developer tool can connect to only one node at a time. If the node in Region_1 fails over to Region_2, the Developer tool cannot connect to the Region_2 node and requires manual reconfiguration of the connection. To enable failover, configure Azure Traffic Manager to provide the connection to whichever node is running.

1. Get the DNS name of the primary VM in each of the two regions.
   The following image shows the location of the DNS name in the **Overview** screen of the VM:



2. In the Azure portal, create a Traffic Manager profile. For the resource group, choose the **Priority** routing method. For the Resource group location, choose Region_1.
3. In the Endpoints configuration screen of the Traffic Manager profile, create an endpoint for the primary VM in Region_1 and the secondary VM in Region_2.
   The following image shows two endpoints created for the Traffic Manager profile:



4. Configure endpoint settings to identify the Traffic Manager profile primary and secondary endpoints. Use **Priority** settings to identify the primary and secondary endpoints.
   a. Configure the following settings for the primary endpoint:

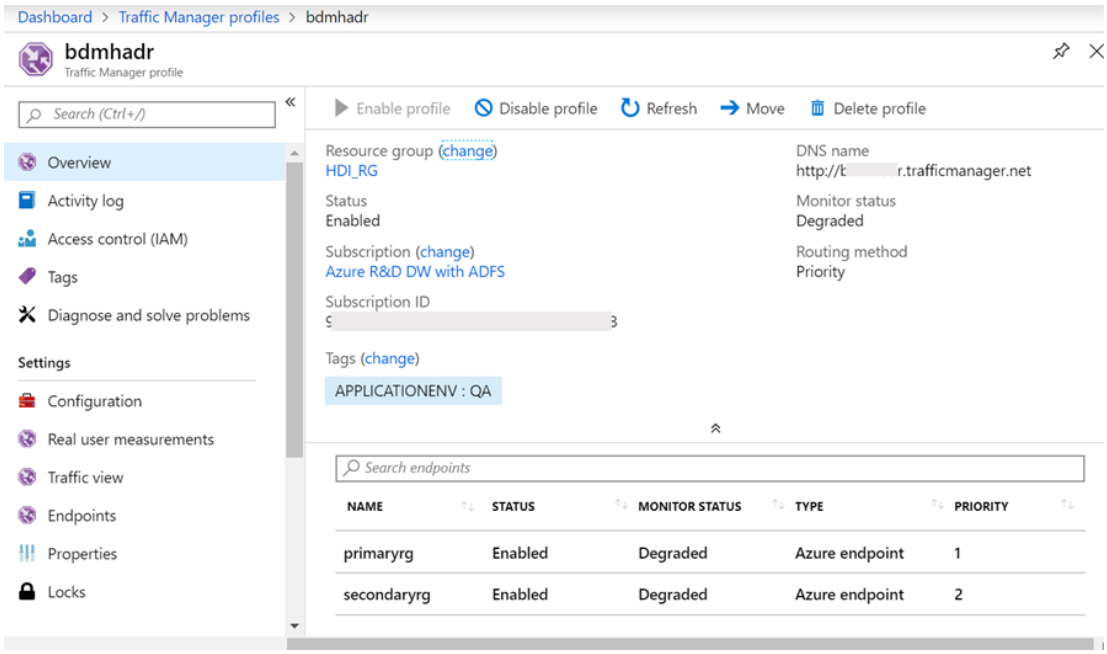
Property	Description	Value
Status	Whether to enable or disable the endpoint.	Select `Enabled`.
Target resource type	Type of the target resource.	Select `IP address`.
Target resource	Name of the target resource VM.	Select the name of the VM in Region_1 where the gateway node resides.
Priority	Priority number for the endpoint.	Select `1`.

   b. Configure the following settings for the secondary endpoint:

Property	Description	Value
Status	Whether to enable or disable the endpoint.	Select `Enabled`.
Target resource type	Type of the target resource.	Select `IP address`.

Property	Description	Value
Target resource	Name of the target resource VM.	Select the name of the VM in Region_2 where the gateway node resides.
Priority	Priority number for the endpoint.	Select 2.

The following image shows an example of a completed Traffic Manager profile:



Note the DNS name in the profile. Supply the DNS name in the next step.

Step 10. Configure Developer Tool Connections with the Traffic Manager Profile

To enable failover in the Developer tool, specify the Azure Traffic Manager profile DNS name as the node that Developer tool connects to.

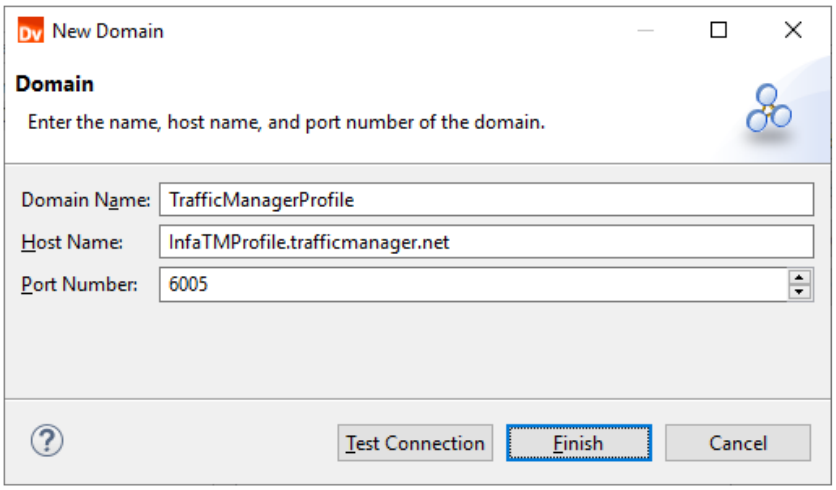
Perform this step in each of the Developer tool instances that connect to the Informatica domains in the high availability system.

1. In the Developer tool, select **File** > **Connect to Repository**.
2. Select **Configure Domains**.
3. Click **Add** to create a new connection, or **Edit** to edit an existing connection.
   The **Edit Domain** dialog box opens.

4. Enter the following properties:

Property	Description	Value
Domain Name	Name of the domain to connect to.	Enter the name of the Traffic Manager profile that you created.
Host Name	Host URL or IP address of the domain to connect to.	Enter the DNS name of the Azure Traffic Manager profile.
Port Number	Port number of the node to connect to.	Enter 6005.

The following image shows an example of the **Edit Domain** dialog box:



5. Optionally click **Test Connection** to test the connection.
6. Click **Finish**.

Step 11. Perform Validation Tests

To validate the system, perform the following tests.

Domain Failover

Perform this test to verify that the Informatica domain fails over to the secondary node in Region_2 when Region_1 fails.

1. Verify that both domain nodes are running nominally.
2. Run a mapping on the Data Integration Service.
3. Manually trigger database failover:
   a. In the Azure portal, browse to the primary database in Region_1.
   b. In the SQL Database blade, select **All settings** > **Geo-Replication**.
   c. In the Secondaries list, select the database you created in Region_2 to be the secondary database.

d. Click the Failover control in the upper right.
The following image shows the Failover control:



Note: Failover takes 1-2 minutes to complete.

4. Verify that the status of the Region_2 database is **Writeable**.
5. Verify that the domain is accessible, and then run the mapping again.

Client Failover

Perform this test to verify that the Developer tool client maintains a connection to the domain when Region_1 fails.

1. Verify that both nodes are running nominally.
2. Verify that the Developer tool can connect to the domain using the Traffic Manager DNS.
3. Perform step 3 in "Domain Failover" to bring down Node 1.
4. Refresh the Developer tool, then run a mapping.

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